

assignment2 - electric filed

 This is a preview of the published version of the quiz

Started: Mar 15 at 3:09pm

Quiz Instructions

2 attempts. Hide the hints first. There are tutorials in my shared folder. I go over the problems step by step.

But try first on your own. And read the Unit about electric field (chapter 21 I think). Work the solved problems.

Also. To check the direction of an electric field use the little yellow sensor here:

https://phet.colorado.edu/sims/html/charges-and-fields/latest/charges-and-fields_en.html 
[\(https://phet.colorado.edu/sims/html/charges-and-fields/latest/charges-and-fields_en.html\)](https://phet.colorado.edu/sims/html/charges-and-fields/latest/charges-and-fields_en.html)

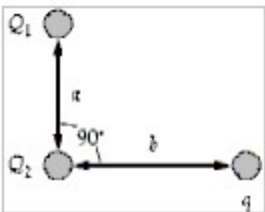
REmember: acceleration = net force / mass and see equation sheet for kinematics

Also. You need to master adding vectors. See tutorials in dropbox.



Question 1 10 pts

If $a = 3.0$ mm, $b = 4.0$ mm, $Q_1 = 60$ nC, $Q_2 = 80$ nC, and $q = 24$ nC in the figure, what is the magnitude of the total electric force on q ?



hint: there is a video_tutorial in my dropbox. See tutorials/ assignment2/ #1

☐

1.9

☐

0.52

☐

2.7N

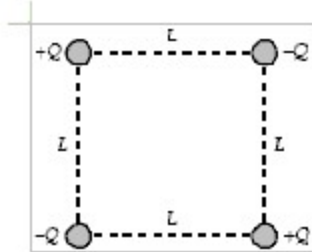


1.5N



Question 2 10 pts

If $Q = 20 \mu\text{C}$ and $L = 60 \text{ cm}$, what is the magnitude of the electrostatic force on any one of the charges shown?



hint: There is a tutorial video in my dropbox. Try first on your own. Its an interesting one.



25N



14N



9.1N

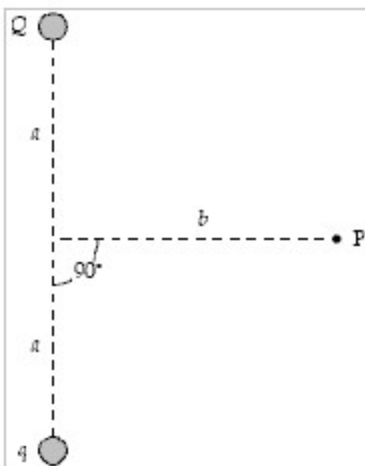


19N



Question 3 10 pts

If $a = 60 \text{ cm}$, $b = 80 \text{ cm}$, $Q = -6.0 \text{ nC}$, and $q = 3.0 \text{ nC}$ in the figure, what is the magnitude of the electric field at point P?



hint: There is a tutorial- video in my dropbox. First, try without watch the video.



67N/C



53N/C



60N/C



71N/C



Question 4 10 pts

A particle (mass = 5.0 g, charge = 40 mC) moves in a region of space where the electric field is uniform and is given by $E_x = -2.3 \text{ N/C}$, $E_y = E_z = 0$. If the position and velocity of the particle at $t = 0$ are given by $x = y = z = 0$ and $v_z = 20 \text{ m/s}$, $v_x = v_y = 0$, what is the distance from the origin to the particle at $t = 2.0 \text{ s}$?

hint: there is a tutorial_video in my shared folder. acceleration = $F_{\text{net}}/\text{mass}$ Use kinematics



54m



60m



3.2m



78m



Question 5 10 pts

Imagine a single charge q placed at one corner of a square, and that the electric field at the center of the square is F/q . If two other equal charges are placed at the adjacent corners of the square (leaving the opposite corner "blank"), the electric field at the center of the square is

hint: its actually easy. Draw.

 F/q . $F/(2q)$.

none of the above

 $F/(4q)$. $4F/q$.

Question 6 10 pts

What is the magnitude of a the vertical electric field that will balance the weight of a plastic sphere of mass 2.1g that has been charged to -3.0 nC? ($k = 1/4\pi\epsilon_0 = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)

hint: this is similar to Milikan experiment. the electric force (qE) is balanced by the weight (mg).
convert nC to C. grams to kg. Slide 66

☐

$6.9 \times 10^6 \text{ N/C}$

☐

$1.5 \times 10^6 \text{ N/C}$

☐

$2.1 \times 10^6 \text{ N/C}$

☐

$7.8 \times 10^5 \text{ N/C}$

⋮

Question 7 10 pts

A proton is placed in an electric field of intensity 800 N/C. What are the magnitude and direction of the acceleration of the proton due to this field? ($e = 1.60 \times 10^{-19} \text{ C}$, $m_{\text{proton}} = 1.67 \times 10^{-27} \text{ kg}$)

acceleration = $F_{\text{net}}/\text{mass}$

☐

$7.66 \times 10^{10} \text{ m/s}^2$ in the direction of the electric field

☐

$7.66 \times 10^9 \text{ m/s}^2$ opposite to the electric field

☐

$76.6 \times 10^{10} \text{ m/s}^2$ opposite to the electric field

☐

$7.66 \times 10^{10} \text{ m/s}^2$ opposite to the electric field

☐

$76.6 \times 10^{10} \text{ m/s}^2$ in the direction of the electric field

⋮

Question 8 10 pts

What is the magnitude of the electric field 2.8 cm from a tiny object that carries an excess charge of -16 nC? ($k = 1/4\pi\epsilon_0 = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)

hint: magnitude is positive. Use the definition. ITs its tiny, its like a point

☐

5100 N/C

 $1.8 \times 10^{14} \text{ N/C}$  $-180,000 \text{ N/C}$  $180,000 \text{ N/C}$  -5100 N/C 

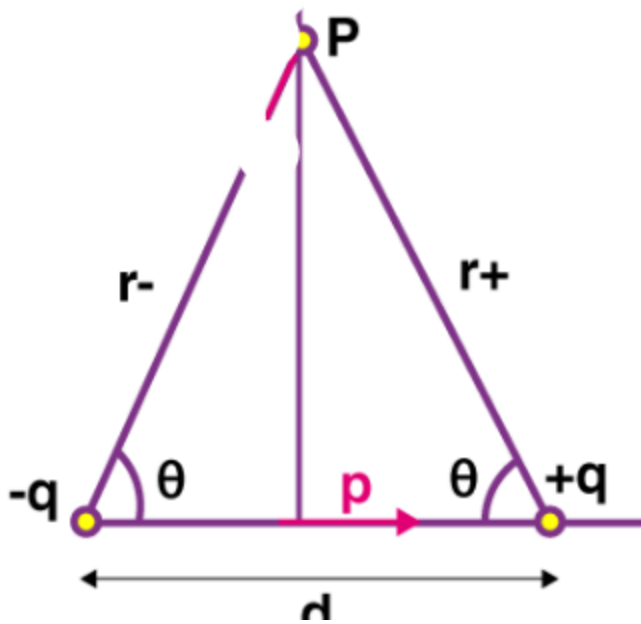
Question 9 10 pts

Two tiny particles having charges $+20.0 \mu\text{C}$ and $-8.00 \mu\text{C}$ are separated by a distance of 20.0 cm . What are the magnitude and direction of electric field midway between these two charges? ($k = 1/4\pi\epsilon_0 = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)

 $25.2 \times 10^5 \text{ N/C}$ directed towards the negative charge $25.2 \times 10^6 \text{ N/C}$ directed towards the positive charge $25.2 \times 10^6 \text{ N/C}$ directed towards the negative charge $25.2 \times 10^5 \text{ N/C}$ directed towards the positive charge $25.2 \times 10^4 \text{ N/C}$ directed towards the negative charge

Question 10 3 pts

You can use the app / add vectors



The electric field at point P is

☐

@ up

☐

0

☐

@ down

☐

@ left

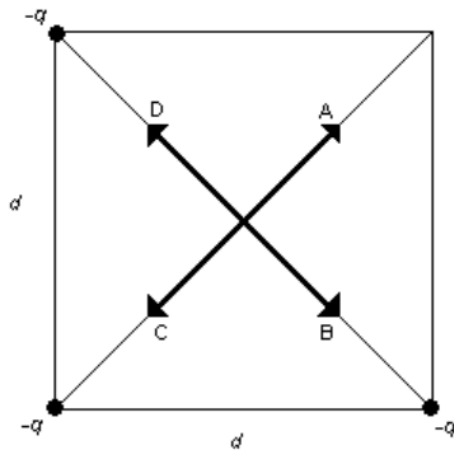
☐

@ right

☐

Question 11 5 pts

) Three equal negative point charges are placed at three of the corners of a square of side d as shown in the figure. Which of the arrows represents the direction of the net electric field at the center of the square?


☐

A

☐

C

☐

E

☐

B

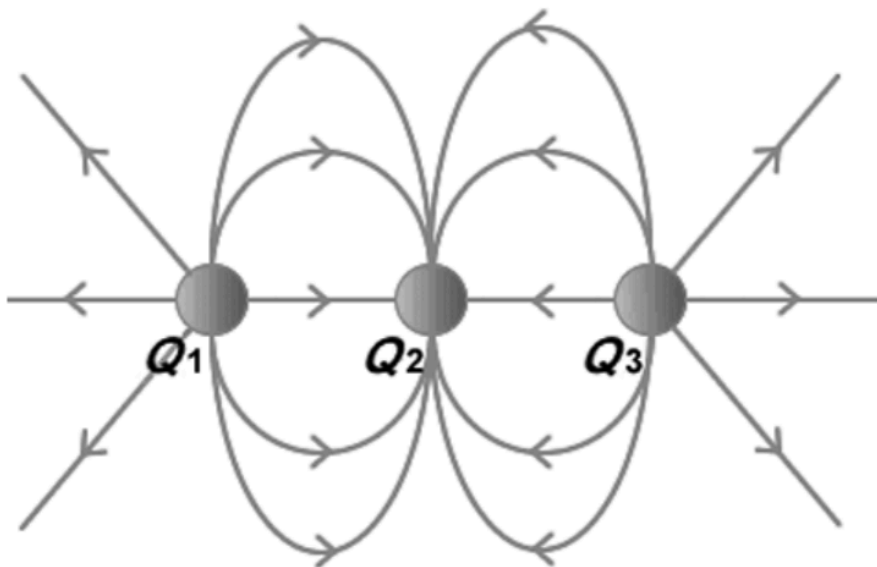
☐

D

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Question 12 5 pts

) The figure shows three electric charges labeled Q_1 , Q_2 , Q_3 , and some electric field lines in the region surrounding the charges. What are the signs of the three charges?



- A) Q_1 is positive, Q_2 is negative, Q_3 is positive.
 B) Q_1 is negative, Q_2 is positive, Q_3 is negative.
 C) Q_1 is positive, Q_2 is positive, Q_3 is negative.
 D) All three charges are negative.
 E) All three charges are positive.

☐

C

☐

B

☐

D

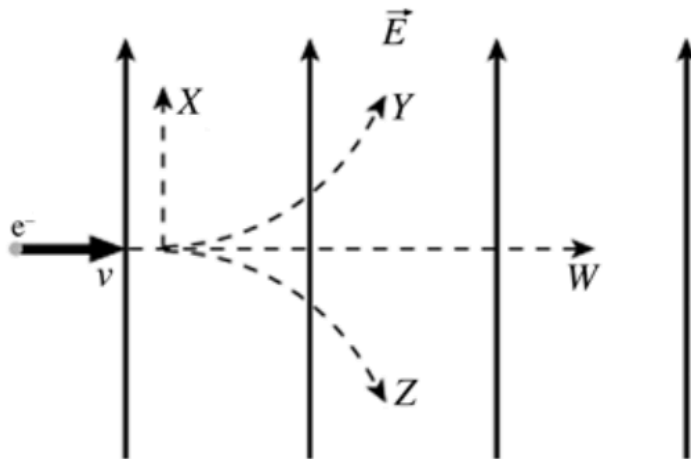
☐

A

☒

Question 13 5 pts

Which trajectory shown below will the electron follow?



- A) trajectory W
 B) trajectory X
 C) trajectory Y
 D) trajectory Z

hint: remember Physics one the electron wants to move in a straight line (inertia) but it cant because there there an electron force (direction?). So along the x-axis its moving at a constant velocity but its also accelerating toward ? when you combine the 2 motions you get.

☐

X



Z



Y



W



Question 14 2 pts

There is an uniform electric field from left to right. (in the place of your screen).

An electron placed in this field



accelerates @ left



moves at a constant speed @ left



accelerates @ right



does not move

Not saved

Submit Quiz